
SOURCES AND USES OF FUNDS

Brecksville-Broadview Heights City School District
Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Dated Date 10/15/2019
Delivery Date 10/15/2019

Sources:

Bond Proceeds:	
Par Amount	43,934,917.40
Premium	7,713,611.60
	51,648,529.00

Uses:

Refunding Escrow Deposits:	
Cash Deposit	0.22
SLGS Purchases	
	51,298,130.00
	51,298,130.22
Delivery Date Expenses:	
Cost of Issuance	350,000.00
Other Uses of Funds:	
Additional Proceeds	398.78
	51,648,529.00

Notes:

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2. The use of the Aa2 rating is consistent with the rating of the outstanding prior bonds.
3. Interest rate assumptions are based on current market conditions and similar credits.
4. Actual results may differ, and Stifel makes no commitments to underwriter at these levels.
5. Costs of issuance and underwriter's discount are estimates for discussion purposes.
6. Analysis was performed with no changes to the term or the structure of the debt service from the currently outstanding issue.
7. Non-interest paying capital appreciation bonds have been included in a manner consistent with the District's existing debt structure solely to generate premium to pay costs of issuance and fund a portion of the escrow to comply with State law requirements restricting the par amount of the refunding bonds.

SUMMARY OF REFUNDING RESULTS

Brecksville-Broadview Heights City School District
Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Dated Date	10/15/2019
Delivery Date	10/15/2019
Arbitrage yield	3.249984%
Escrow yield	1.397453%
Value of Negative Arbitrage	3,389,333.59
Bond Par Amount	43,934,917.40
True Interest Cost	3.252709%
Average Life	25.269
Par amount of refunded bonds	43,960,000.00
Average coupon of refunded bonds	5.123125%
Average life of refunded bonds	26.344
PV of prior debt to 10/15/2019 @ 3.249984%	58,965,554.67
Net PV Savings	7,294,512.37
Percentage savings of refunded bonds	16.593522%

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BOND PRICING

Brecksville-Broadview Heights City School District
Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Bond Component	Maturity Date	Amount	Rate	Yield	Price	Yield to Maturity	Call Date	Call Price	Principal per \$5,000 at Maturity
Serial Bonds:									
	12/01/2020	310,000.00	4.000%	2.050%	102.16100000				
	12/01/2021	505,000.00	4.000%	1.930%	104.29300000				
	12/01/2022	525,000.00	4.000%	1.890%	106.37700000				
	12/01/2023	550,000.00	4.000%	1.900%	108.29700000				
	12/01/2024	570,000.00	4.000%	2.000%	109.69900000				
	12/01/2025	590,000.00	4.000%	2.060%	111.11200000				
	12/01/2026	615,000.00	4.000%	2.160%	112.09300000				
	12/01/2027	640,000.00	4.000%	2.250%	112.93000000				
	12/01/2028	665,000.00	4.000%	2.300%	112.53400000	C	2.458%	12/01/2027	100.000
	12/01/2029	695,000.00	4.000%	2.400%	111.74800000	C	2.668%	12/01/2027	100.000
	12/01/2030	700,000.00	4.000%	2.500%	110.96800000	C	2.843%	12/01/2027	100.000
		<u>6,365,000.00</u>							
Term Bond:									
	12/01/2041	5,200,000.00	3.070%	3.070%	100.00000000				
Term Bond:									
	12/01/2047	12,730,000.00	3.170%	3.170%	100.00000000				
Term Bond:									
	12/01/2054	19,425,000.00	3.270%	3.270%	100.00000000				
CAB Bond:									
	12/01/2030	28,301.95	30.000%	3.150%	1,584.56360781				222.85
	12/01/2031	49,539.00	30.000%	3.300%	1,995.13353116				168.50
	12/01/2032	37,583.00	30.000%	3.450%	2,504.82731554				127.40
	12/01/2033	28,712.30	30.000%	3.600%	3,134.71717696				96.35
	12/01/2034	22,583.50	30.000%	3.750%	3,912.42278655				72.85
	12/01/2035	17,065.50	30.000%	3.900%	4,871.66212534				55.05
	12/01/2036	13,119.75	30.000%	4.070%	6,020.64825930				41.65
	12/01/2037	10,206.00	30.000%	4.120%	7,578.57142857				31.50
	12/01/2038	7,806.40	30.000%	4.170%	9,539.91596639				23.80
		<u>214,917.40</u>							
		43,934,917.40							

BOND PRICING

Brecksville-Broadview Heights City School District
Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Dated Date	10/15/2019	
Delivery Date	10/15/2019	
First Coupon	06/01/2020	
Par Amount	43,934,917.40	
Premium	7,713,611.60	
Production	51,648,529.00	117.556905%
Underwriter's Discount		
Purchase Price	51,648,529.00	117.556905%
Accrued Interest		
Net Proceeds	51,648,529.00	

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BOND SUMMARY STATISTICS

Brecksville-Broadview Heights City School District Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Dated Date	10/15/2019
Delivery Date	10/15/2019
Last Maturity	12/01/2054
Arbitrage Yield	3.249984%
True Interest Cost (TIC)	3.252709%
All-In TIC	3.294549%
Average Life (years)	25.269
Duration of Issue (years)	16.538
Par Amount	43,934,917.40
Bond Proceeds	51,648,529.00
Total Interest	35,966,018.36
Net Interest	28,252,406.76
Total Debt Service	92,691,018.36
Maximum Annual Debt Service	3,172,426.50
Average Annual Debt Service	2,638,681.53
Underwriter's Fees (per \$1000)	
Average Takedown	
Other Fee	
Total Underwriter's Discount	
Bid Price	117.556905

Bond Component	Par Value	Price	Average Coupon	Average Life	PV of 1 bp change
Serial Bonds	6,365,000.00	109.854	4.000%	6.655	3,734.45
CAB Bond	214,917.40	3,397.255		13.915	10,838.85
Term Bond	5,200,000.00	100.000	3.070%	21.164	8,268.00
Term Bond	12,730,000.00	100.000	3.170%	25.760	23,550.50
Term Bond	19,425,000.00	100.000	3.270%	32.270	40,404.00
	43,934,917.40			25.269	86,795.80

	TIC	All-In TIC	Arbitrage Yield
Par Value	43,934,917.40	43,934,917.40	43,934,917.40
+ Accrued Interest			
+ Premium (Discount)	7,713,611.60	7,713,611.60	7,713,611.60
- Underwriter's Discount			
- Cost of Issuance Expense		-350,000.00	
- Other Amounts			
Target Value	51,648,529.00	51,298,529.00	51,648,529.00
Target Date	10/15/2019	10/15/2019	10/15/2019
Yield	3.252709%	3.294549%	3.249984%

BOND SUMMARY STATISTICS

Brecksville-Broadview Heights City School District Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

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BOND DEBT SERVICE

**Brecksville-Broadview Heights City School District
Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018**

Period Ending	Principal	Coupon	Interest	Compounded Interest	Debt Service	Annual Debt Service
06/01/2020			912,147.61		912,147.61	
12/01/2020	310,000.00	4.000%	726,489.25		1,036,489.25	1,948,636.86
06/01/2021			720,289.25		720,289.25	
12/01/2021	505,000.00	4.000%	720,289.25		1,225,289.25	1,945,578.50
06/01/2022			710,189.25		710,189.25	
12/01/2022	525,000.00	4.000%	710,189.25		1,235,189.25	1,945,378.50
06/01/2023			699,689.25		699,689.25	
12/01/2023	550,000.00	4.000%	699,689.25		1,249,689.25	1,949,378.50
06/01/2024			688,689.25		688,689.25	
12/01/2024	570,000.00	4.000%	688,689.25		1,258,689.25	1,947,378.50
06/01/2025			677,289.25		677,289.25	
12/01/2025	590,000.00	4.000%	677,289.25		1,267,289.25	1,944,578.50
06/01/2026			665,489.25		665,489.25	
12/01/2026	615,000.00	4.000%	665,489.25		1,280,489.25	1,945,978.50
06/01/2027			653,189.25		653,189.25	
12/01/2027	640,000.00	4.000%	653,189.25		1,293,189.25	1,946,378.50
06/01/2028			640,389.25		640,389.25	
12/01/2028	665,000.00	4.000%	640,389.25		1,305,389.25	1,945,778.50
06/01/2029			627,089.25		627,089.25	
12/01/2029	695,000.00	4.000%	627,089.25		1,322,089.25	1,949,178.50
06/01/2030			613,189.25		613,189.25	
12/01/2030	728,301.95	** %	613,189.25	606,698.05	1,948,189.25	2,561,378.50
06/01/2031			599,189.25		599,189.25	
12/01/2031	49,539.00	30.000%	599,189.25	1,420,461.00	2,069,189.25	2,668,378.50
06/01/2032			599,189.25		599,189.25	
12/01/2032	37,583.00	30.000%	599,189.25	1,437,417.00	2,074,189.25	2,673,378.50
06/01/2033			599,189.25		599,189.25	
12/01/2033	28,712.30	30.000%	599,189.25	1,461,287.70	2,089,189.25	2,688,378.50
06/01/2034			599,189.25		599,189.25	
12/01/2034	22,583.50	30.000%	599,189.25	1,527,416.50	2,149,189.25	2,748,378.50
06/01/2035			599,189.25		599,189.25	
12/01/2035	17,065.50	30.000%	599,189.25	1,532,934.50	2,149,189.25	2,748,378.50
06/01/2036			599,189.25		599,189.25	
12/01/2036	13,119.75	30.000%	599,189.25	1,561,880.25	2,174,189.25	2,773,378.50
06/01/2037			599,189.25		599,189.25	
12/01/2037	10,206.00	30.000%	599,189.25	1,609,794.00	2,219,189.25	2,818,378.50
06/01/2038			599,189.25		599,189.25	
12/01/2038	7,806.40	30.000%	599,189.25	1,632,193.60	2,239,189.25	2,838,378.50
06/01/2039			599,189.25		599,189.25	
12/01/2039	1,625,000.00	3.070%	599,189.25		2,224,189.25	2,823,378.50
06/01/2040			574,245.50		574,245.50	
12/01/2040	1,760,000.00	3.070%	574,245.50		2,334,245.50	2,908,491.00
06/01/2041			547,229.50		547,229.50	
12/01/2041	1,815,000.00	3.070%	547,229.50		2,362,229.50	2,909,459.00
06/01/2042			519,369.25		519,369.25	
12/01/2042	1,870,000.00	3.170%	519,369.25		2,389,369.25	2,908,738.50
06/01/2043			489,729.75		489,729.75	
12/01/2043	2,000,000.00	3.170%	489,729.75		2,489,729.75	2,979,459.50
06/01/2044			458,029.75		458,029.75	
12/01/2044	2,075,000.00	3.170%	458,029.75		2,533,029.75	2,991,059.50
06/01/2045			425,141.00		425,141.00	
12/01/2045	2,150,000.00	3.170%	425,141.00		2,575,141.00	3,000,282.00
06/01/2046			391,063.50		391,063.50	
12/01/2046	2,270,000.00	3.170%	391,063.50		2,661,063.50	3,052,127.00
06/01/2047			355,084.00		355,084.00	

BOND DEBT SERVICE

**Brecksville-Broadview Heights City School District
Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018**

Period Ending	Principal	Coupon	Interest	Compounded Interest	Debt Service	Annual Debt Service
12/01/2047	2,365,000.00	3.170%	355,084.00		2,720,084.00	3,075,168.00
06/01/2048			317,598.75		317,598.75	
12/01/2048	2,450,000.00	3.270%	317,598.75		2,767,598.75	3,085,197.50
06/01/2049			277,541.25		277,541.25	
12/01/2049	2,590,000.00	3.270%	277,541.25		2,867,541.25	3,145,082.50
06/01/2050			235,194.75		235,194.75	
12/01/2050	2,690,000.00	3.270%	235,194.75		2,925,194.75	3,160,389.50
06/01/2051			191,213.25		191,213.25	
12/01/2051	2,790,000.00	3.270%	191,213.25		2,981,213.25	3,172,426.50
06/01/2052			145,596.75		145,596.75	
12/01/2052	2,880,000.00	3.270%	145,596.75		3,025,596.75	3,171,193.50
06/01/2053			98,508.75		98,508.75	
12/01/2053	2,970,000.00	3.270%	98,508.75		3,068,508.75	3,167,017.50
06/01/2054			49,949.25		49,949.25	
12/01/2054	3,055,000.00	3.270%	49,949.25		3,104,949.25	3,154,898.50
	43,934,917.40		35,966,018.36	12,790,082.60	92,691,018.36	92,691,018.36

Notes:

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SAVINGS

Brecksville-Broadview Heights City School District
Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Date	Prior Debt Service	Refunding Debt Service	Savings	Annual Savings	Present Value to 10/15/2019 @ 3.2499835%
12/01/2019	1,120,750.00		1,120,750.00	1,120,750.00	1,116,142.70
06/01/2020	1,120,750.00	912,147.61	208,602.39		204,422.99
12/01/2020	1,120,750.00	1,036,489.25	84,260.75	292,863.14	81,252.22
06/01/2021	1,120,750.00	720,289.25	400,460.75		379,987.55
12/01/2021	1,120,750.00	1,225,289.25	-104,539.25	295,921.50	-97,608.64
06/01/2022	1,120,750.00	710,189.25	410,560.75		377,212.25
12/01/2022	1,120,750.00	1,235,189.25	-114,439.25	296,121.50	-103,462.47
06/01/2023	1,120,750.00	699,689.25	421,060.75		374,586.46
12/01/2023	1,120,750.00	1,249,689.25	-128,939.25	292,121.50	-112,873.48
06/01/2024	1,120,750.00	688,689.25	432,060.75		372,178.33
12/01/2024	1,120,750.00	1,258,689.25	-137,939.25	294,121.50	-116,921.29
06/01/2025	1,120,750.00	677,289.25	443,460.75		369,879.62
12/01/2025	1,120,750.00	1,267,289.25	-146,539.25	296,921.50	-120,270.37
06/01/2026	1,120,750.00	665,489.25	455,260.75		367,675.24
12/01/2026	1,120,750.00	1,280,489.25	-159,739.25	295,521.50	-126,944.91
06/01/2027	1,120,750.00	653,189.25	467,560.75		365,629.45
12/01/2027	1,120,750.00	1,293,189.25	-172,439.25	295,121.50	-132,690.16
06/01/2028	1,120,750.00	640,389.25	480,360.75		363,722.02
12/01/2028	1,120,750.00	1,305,389.25	-184,639.25	295,721.50	-137,570.58
06/01/2029	1,120,750.00	627,089.25	493,660.75		361,934.21
12/01/2029	1,120,750.00	1,322,089.25	-201,339.25	292,321.50	-145,254.29
06/01/2030	1,120,750.00	613,189.25	507,560.75		360,319.71
12/01/2030	1,735,750.00	1,948,189.25	-212,439.25	295,121.50	-148,400.11
06/01/2031	1,105,375.00	599,189.25	506,185.75		347,943.60
12/01/2031	1,855,375.00	2,069,189.25	-213,814.25	292,371.50	-144,622.23
06/01/2032	1,086,625.00	599,189.25	487,435.75		324,425.71
12/01/2032	1,881,625.00	2,074,189.25	-192,564.25	294,871.50	-126,116.83
06/01/2033	1,066,750.00	599,189.25	467,560.75		301,324.81
12/01/2033	1,916,750.00	2,089,189.25	-172,439.25	295,121.50	-109,353.44
06/01/2034	1,045,500.00	599,189.25	446,310.75		278,505.10
12/01/2034	1,995,500.00	2,149,189.25	-153,689.25	292,621.50	-94,371.05
06/01/2035	1,021,750.00	599,189.25	422,560.75		255,319.45
12/01/2035	2,021,750.00	2,149,189.25	-127,439.25	295,121.50	-75,770.03
06/01/2036	996,750.00	599,189.25	397,560.75		232,593.30
12/01/2036	2,071,750.00	2,174,189.25	-102,439.25	295,121.50	-58,973.86
06/01/2037	969,875.00	599,189.25	370,685.75		209,989.97
12/01/2037	2,144,875.00	2,219,189.25	-74,314.25	296,371.50	-41,425.16
06/01/2038	940,500.00	599,189.25	341,310.75		187,215.40
12/01/2038	2,190,500.00	2,239,189.25	-48,689.25	292,621.50	-26,279.93
06/01/2039	909,250.00	599,189.25	310,060.75		164,678.67
12/01/2039	2,209,250.00	2,224,189.25	-14,939.25	295,121.50	-7,807.62
06/01/2040	876,750.00	574,245.50	302,504.50		155,568.39
12/01/2040	2,326,750.00	2,334,245.50	-7,495.50	295,009.00	-3,793.06
06/01/2041	840,500.00	547,229.50	293,270.50		146,034.96
12/01/2041	2,365,500.00	2,362,229.50	3,270.50	296,541.00	1,602.51
06/01/2042	802,375.00	519,369.25	283,005.75		136,452.87
12/01/2042	2,402,375.00	2,389,369.25	13,005.75	296,011.50	6,170.53
06/01/2043	762,375.00	489,729.75	272,645.25		127,287.08
12/01/2043	2,512,375.00	2,489,729.75	22,645.25	295,290.50	10,403.10
06/01/2044	718,625.00	458,029.75	260,595.25		117,801.77
12/01/2044	2,568,625.00	2,533,029.75	35,595.25	296,190.50	15,833.50
06/01/2045	672,375.00	425,141.00	247,234.00		108,216.25
12/01/2045	2,622,375.00	2,575,141.00	47,234.00	294,468.00	20,344.10
06/01/2046	623,625.00	391,063.50	232,561.50		98,564.62

SAVINGS

Brecksville-Broadview Heights City School District Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Date	Prior Debt Service	Refunding Debt Service	Savings	Annual Savings	Present Value to 10/15/2019 @ 3.2499835%
12/01/2046	2,723,625.00	2,661,063.50	62,561.50	295,123.00	26,090.95
06/01/2047	571,125.00	355,084.00	216,041.00		88,658.09
12/01/2047	2,796,125.00	2,720,084.00	76,041.00	292,082.00	30,706.44
06/01/2048	515,500.00	317,598.75	197,901.25		78,637.50
12/01/2048	2,865,500.00	2,767,598.75	97,901.25	295,802.50	38,279.73
06/01/2049	456,750.00	277,541.25	179,208.75		68,950.80
12/01/2049	2,981,750.00	2,867,541.25	114,208.75	293,417.50	43,239.33
06/01/2050	390,468.75	235,194.75	155,274.00		57,846.59
12/01/2050	3,065,468.75	2,925,194.75	140,274.00	295,548.00	51,422.80
06/01/2051	320,250.00	191,213.25	129,036.75		46,546.97
12/01/2051	3,145,250.00	2,981,213.25	164,036.75	293,073.50	58,226.22
06/01/2052	246,093.75	145,596.75	100,497.00		35,101.85
12/01/2052	3,221,093.75	3,025,596.75	195,497.00	295,994.00	67,191.84
06/01/2053	168,000.00	98,508.75	69,491.25		23,502.07
12/01/2053	3,293,000.00	3,068,508.75	224,491.25	293,982.50	74,709.32
06/01/2054	85,968.75	49,949.25	36,019.50		11,795.40
12/01/2054	3,360,968.75	3,104,949.25	256,019.50	292,039.00	82,498.75
	104,123,562.50	92,691,018.36	11,432,544.14	11,432,544.14	7,294,113.59

Savings Summary

PV of savings from cash flow	7,294,113.59
Plus: Refunding funds on hand	398.78
Net PV Savings	7,294,512.37

Notes:

1. Preliminary and subject to change.
2. The use of the Aa2 rating is consistent with the rating of the outstanding prior bonds.
3. Interest rate assumptions are based on current market conditions and similar credits.
4. Actual results may differ, and Stifel makes no commitments to underwriter at these levels.
5. Costs of issuance and underwriter's discount are estimates for discussion purposes.
6. Analysis was performed with no changes to the term or the structure of the debt service from the currently outstanding issue.
7. Non-interest paying capital appreciation bonds have been included in a manner consistent with the District's existing debt structure solely to generate premium to pay costs of issuance and fund a portion of the escrow to comply with State law requirements restricting the par amount of the refunding bonds.

SAVINGS

Brecksville-Broadview Heights City School District Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Date	Prior Debt Service	Refunding Debt Service	Savings	Present Value to 10/15/2019 @ 3.2499835%
12/01/2019	1,120,750.00		1,120,750.00	1,116,142.70
12/01/2020	2,241,500.00	1,948,636.86	292,863.14	285,675.21
12/01/2021	2,241,500.00	1,945,578.50	295,921.50	282,378.91
12/01/2022	2,241,500.00	1,945,378.50	296,121.50	273,749.78
12/01/2023	2,241,500.00	1,949,378.50	292,121.50	261,712.97
12/01/2024	2,241,500.00	1,947,378.50	294,121.50	255,257.04
12/01/2025	2,241,500.00	1,944,578.50	296,921.50	249,609.25
12/01/2026	2,241,500.00	1,945,978.50	295,521.50	240,730.33
12/01/2027	2,241,500.00	1,946,378.50	295,121.50	232,939.29
12/01/2028	2,241,500.00	1,945,778.50	295,721.50	226,151.43
12/01/2029	2,241,500.00	1,949,178.50	292,321.50	216,679.92
12/01/2030	2,856,500.00	2,561,378.50	295,121.50	211,919.60
12/01/2031	2,960,750.00	2,668,378.50	292,371.50	203,321.37
12/01/2032	2,968,250.00	2,673,378.50	294,871.50	198,308.89
12/01/2033	2,983,500.00	2,688,378.50	295,121.50	191,971.37
12/01/2034	3,041,000.00	2,748,378.50	292,621.50	184,134.05
12/01/2035	3,043,500.00	2,748,378.50	295,121.50	179,549.43
12/01/2036	3,068,500.00	2,773,378.50	295,121.50	173,619.44
12/01/2037	3,114,750.00	2,818,378.50	296,371.50	168,564.80
12/01/2038	3,131,000.00	2,838,378.50	292,621.50	160,935.47
12/01/2039	3,118,500.00	2,823,378.50	295,121.50	156,871.05
12/01/2040	3,203,500.00	2,908,491.00	295,009.00	151,775.33
12/01/2041	3,206,000.00	2,909,459.00	296,541.00	147,637.48
12/01/2042	3,204,750.00	2,908,738.50	296,011.50	142,623.40
12/01/2043	3,274,750.00	2,979,459.50	295,290.50	137,690.18
12/01/2044	3,287,250.00	2,991,059.50	296,190.50	133,635.27
12/01/2045	3,294,750.00	3,000,282.00	294,468.00	128,560.35
12/01/2046	3,347,250.00	3,052,127.00	295,123.00	124,655.57
12/01/2047	3,367,250.00	3,075,168.00	292,082.00	119,364.53
12/01/2048	3,381,000.00	3,085,197.50	295,802.50	116,917.23
12/01/2049	3,438,500.00	3,145,082.50	293,417.50	112,190.13
12/01/2050	3,455,937.50	3,160,389.50	295,548.00	109,269.39
12/01/2051	3,465,500.00	3,172,426.50	293,073.50	104,773.19
12/01/2052	3,467,187.50	3,171,193.50	295,994.00	102,293.69
12/01/2053	3,461,000.00	3,167,017.50	293,982.50	98,211.39
12/01/2054	3,446,937.50	3,154,898.50	292,039.00	94,294.15
	104,123,562.50	92,691,018.36	11,432,544.14	7,294,113.59

Savings Summary

PV of savings from cash flow	7,294,113.59
Plus: Refunding funds on hand	398.78
Net PV Savings	7,294,512.37

SAVINGS

Brecksville-Broadview Heights City School District
Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Notes:

1. Preliminary and subject to change.
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7. Non-interest paying capital appreciation bonds have been included in a manner consistent with the District's existing debt structure solely to generate premium to pay costs of issuance and fund a portion of the escrow to comply with State law requirements restricting the par amount of the refunding bonds.

PRIOR BOND DEBT SERVICE

Brecksville-Broadview Heights City School District
Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Period Ending	Principal	Coupon	Interest	Debt Service	Annual Debt Service
12/01/2019			1,120,750.00	1,120,750.00	1,120,750.00
06/01/2020			1,120,750.00	1,120,750.00	
12/01/2020			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2021			1,120,750.00	1,120,750.00	
12/01/2021			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2022			1,120,750.00	1,120,750.00	
12/01/2022			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2023			1,120,750.00	1,120,750.00	
12/01/2023			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2024			1,120,750.00	1,120,750.00	
12/01/2024			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2025			1,120,750.00	1,120,750.00	
12/01/2025			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2026			1,120,750.00	1,120,750.00	
12/01/2026			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2027			1,120,750.00	1,120,750.00	
12/01/2027			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2028			1,120,750.00	1,120,750.00	
12/01/2028			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2029			1,120,750.00	1,120,750.00	
12/01/2029			1,120,750.00	1,120,750.00	2,241,500.00
06/01/2030			1,120,750.00	1,120,750.00	
12/01/2030	615,000	5.000%	1,120,750.00	1,735,750.00	2,856,500.00
06/01/2031			1,105,375.00	1,105,375.00	
12/01/2031	750,000	5.000%	1,105,375.00	1,855,375.00	2,960,750.00
06/01/2032			1,086,625.00	1,086,625.00	
12/01/2032	795,000	5.000%	1,086,625.00	1,881,625.00	2,968,250.00
06/01/2033			1,066,750.00	1,066,750.00	
12/01/2033	850,000	5.000%	1,066,750.00	1,916,750.00	2,983,500.00
06/01/2034			1,045,500.00	1,045,500.00	
12/01/2034	950,000	5.000%	1,045,500.00	1,995,500.00	3,041,000.00
06/01/2035			1,021,750.00	1,021,750.00	
12/01/2035	1,000,000	5.000%	1,021,750.00	2,021,750.00	3,043,500.00
06/01/2036			996,750.00	996,750.00	
12/01/2036	1,075,000	5.000%	996,750.00	2,071,750.00	3,068,500.00
06/01/2037			969,875.00	969,875.00	
12/01/2037	1,175,000	5.000%	969,875.00	2,144,875.00	3,114,750.00
06/01/2038			940,500.00	940,500.00	
12/01/2038	1,250,000	5.000%	940,500.00	2,190,500.00	3,131,000.00
06/01/2039			909,250.00	909,250.00	
12/01/2039	1,300,000	5.000%	909,250.00	2,209,250.00	3,118,500.00
06/01/2040			876,750.00	876,750.00	
12/01/2040	1,450,000	5.000%	876,750.00	2,326,750.00	3,203,500.00
06/01/2041			840,500.00	840,500.00	
12/01/2041	1,525,000	5.000%	840,500.00	2,365,500.00	3,206,000.00
06/01/2042			802,375.00	802,375.00	
12/01/2042	1,600,000	5.000%	802,375.00	2,402,375.00	3,204,750.00
06/01/2043			762,375.00	762,375.00	
12/01/2043	1,750,000	5.000%	762,375.00	2,512,375.00	3,274,750.00
06/01/2044			718,625.00	718,625.00	
12/01/2044	1,850,000	5.000%	718,625.00	2,568,625.00	3,287,250.00
06/01/2045			672,375.00	672,375.00	
12/01/2045	1,950,000	5.000%	672,375.00	2,622,375.00	3,294,750.00
06/01/2046			623,625.00	623,625.00	
12/01/2046	2,100,000	5.000%	623,625.00	2,723,625.00	3,347,250.00

PRIOR BOND DEBT SERVICE

Brecksville-Broadview Heights City School District
Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Period Ending	Principal	Coupon	Interest	Debt Service	Annual Debt Service
06/01/2047			571,125.00	571,125.00	
12/01/2047	2,225,000	5.000%	571,125.00	2,796,125.00	3,367,250.00
06/01/2048			515,500.00	515,500.00	
12/01/2048	2,350,000	5.000%	515,500.00	2,865,500.00	3,381,000.00
06/01/2049			456,750.00	456,750.00	
12/01/2049	2,525,000	5.250%	456,750.00	2,981,750.00	3,438,500.00
06/01/2050			390,468.75	390,468.75	
12/01/2050	2,675,000	5.250%	390,468.75	3,065,468.75	3,455,937.50
06/01/2051			320,250.00	320,250.00	
12/01/2051	2,825,000	5.250%	320,250.00	3,145,250.00	3,465,500.00
06/01/2052			246,093.75	246,093.75	
12/01/2052	2,975,000	5.250%	246,093.75	3,221,093.75	3,467,187.50
06/01/2053			168,000.00	168,000.00	
12/01/2053	3,125,000	5.250%	168,000.00	3,293,000.00	3,461,000.00
06/01/2054			85,968.75	85,968.75	
12/01/2054	3,275,000	5.250%	85,968.75	3,360,968.75	3,446,937.50
	43,960,000		60,163,562.50	104,123,562.50	104,123,562.50

Notes:

1. Preliminary and subject to change.
2. The use of the Aa2 rating is consistent with the rating of the outstanding prior bonds.
3. Interest rate assumptions are based on current market conditions and similar credits.
4. Actual results may differ, and Stifel makes no commitments to underwriter at these levels.
5. Costs of issuance and underwriter's discount are estimates for discussion purposes.
6. Analysis was performed with no changes to the term or the structure of the debt service from the currently outstanding issue.
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SUMMARY OF BONDS REFUNDED

Brecksville-Broadview Heights City School District Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Bond	Maturity Date	Interest Rate	Par Amount	Call Date	Call Price
School Facilities Improvement Bonds, Series 2018, 2018:					
SERIAL	12/01/2030	5.000%	615,000.00	12/01/2023	100.000
	12/01/2031	5.000%	750,000.00	12/01/2023	100.000
	12/01/2032	5.000%	795,000.00	12/01/2023	100.000
	12/01/2033	5.000%	850,000.00	12/01/2023	100.000
	12/01/2034	5.000%	950,000.00	12/01/2023	100.000
	12/01/2035	5.000%	1,000,000.00	12/01/2023	100.000
	12/01/2036	5.000%	1,075,000.00	12/01/2023	100.000
	12/01/2037	5.000%	1,175,000.00	12/01/2023	100.000
	12/01/2038	5.000%	1,250,000.00	12/01/2023	100.000
	12/01/2045	5.000%	11,425,000.00	12/01/2023	100.000
TERM45	12/01/2045	5.000%	11,425,000.00	12/01/2023	100.000
TERM48	12/01/2048	5.000%	6,675,000.00	12/01/2023	100.000
TERM54	12/01/2054	5.250%	17,400,000.00	12/01/2023	100.000
			43,960,000.00		

Notes:

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3. Interest rate assumptions are based on current market conditions and similar credits.
4. Actual results may differ, and Stifel makes no commitments to underwriter at these levels.
5. Costs of issuance and underwriter's discount are estimates for discussion purposes.
6. Analysis was performed with no changes to the term or the structure of the debt service from the currently outstanding issue.
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SUMMARY OF UNREFUNDED BONDS

Brecksville-Broadview Heights City School District Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Bond	Maturity Date	Interest Rate	Par Amount	Value on Oct 15, 2019
School Facilities Improvement Bonds, Series 2018, 2018:				
SERIAL	12/01/2020	2.000%	5,000.00	5,000.00
	12/01/2021	2.000%	5,000.00	5,000.00
CAB	12/01/2022	29.924%	113,985.90	154,663.70
	12/01/2023	29.924%	94,405.50	128,097.45
	12/01/2024	29.924%	78,484.65	106,492.95
	12/01/2025	29.924%	74,732.00	101,399.20
	12/01/2026	29.925%	57,048.05	77,405.00
	12/01/2027	29.924%	43,930.00	59,604.50
	12/01/2028	29.926%	36,703.00	49,803.05
	12/01/2029	29.924%	28,868.40	39,171.00
	12/01/2030	29.926%	1,819.95	2,469.50
			539,977.45	729,106.35

Notes:

1. Preliminary and subject to change.
2. The use of the Aa2 rating is consistent with the rating of the outstanding prior bonds.
3. Interest rate assumptions are based on current market conditions and similar credits.
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5. Costs of issuance and underwriter's discount are estimates for discussion purposes.
6. Analysis was performed with no changes to the term or the structure of the debt service from the currently outstanding issue.
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UNREFUNDED BOND DEBT SERVICE

Brecksville-Broadview Heights City School District
Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Period Ending	Principal	Coupon	Interest	Compounded Interest	Debt Service	Annual Debt Service
12/01/2019			100		100	100
06/01/2020			100		100	
12/01/2020	5,000.00	2.00000000%	100		5,100	5,200
06/01/2021			50		50	
12/01/2021	5,000.00	2.00000000%	50		5,050	5,100
12/01/2022	113,985.90	29.92419306%		256,014.10	370,000	370,000
12/01/2023	94,405.50	29.92408076%		310,594.50	405,000	405,000
12/01/2024	78,484.65	29.92444800%		366,515.35	445,000	445,000
12/01/2025	74,732.00	29.92429490%		485,268.00	560,000	560,000
12/01/2026	57,048.05	29.92482568%		507,951.95	565,000	565,000
12/01/2027	43,930.00	29.92445992%		531,070.00	575,000	575,000
12/01/2028	36,703.00	29.92588836%		598,297.00	635,000	635,000
12/01/2029	28,868.40	29.92431514%		631,131.60	660,000	660,000
12/01/2030	1,819.95	29.92587886%		53,180.05	55,000	55,000
	539,977.45		400	3,740,022.55	4,280,400	4,280,400

Notes:

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ESCROW REQUIREMENTS

Brecksville-Broadview Heights City School District Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Period Ending	Interest	Principal Redeemed	Total
12/01/2019	1,120,750.00		1,120,750.00
06/01/2020	1,120,750.00		1,120,750.00
12/01/2020	1,120,750.00		1,120,750.00
06/01/2021	1,120,750.00		1,120,750.00
12/01/2021	1,120,750.00		1,120,750.00
06/01/2022	1,120,750.00		1,120,750.00
12/01/2022	1,120,750.00		1,120,750.00
06/01/2023	1,120,750.00		1,120,750.00
12/01/2023	1,120,750.00	43,960,000.00	45,080,750.00
	10,086,750.00	43,960,000.00	54,046,750.00

Notes:

1. Preliminary and subject to change.
2. The use of the Aa2 rating is consistent with the rating of the outstanding prior bonds.
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ESCROW DESCRIPTIONS

Brecksville-Broadview Heights City School District Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Type of Security	Type of SLGS	Maturity Date	First Int Pmt Date	Par Amount	Rate	Max Rate
Oct 15, 2019:						
SLGS	Certificate	12/01/2019	12/01/2019	1,028,885	2.050%	2.050%
SLGS	Certificate	06/01/2020	06/01/2020	764,670	1.860%	1.860%
SLGS	Note	12/01/2020	12/01/2019	773,609	1.720%	1.720%
SLGS	Note	06/01/2021	12/01/2019	780,261	1.600%	1.600%
SLGS	Note	12/01/2021	12/01/2019	786,503	1.510%	1.510%
SLGS	Note	06/01/2022	12/01/2019	792,441	1.460%	1.460%
SLGS	Note	12/01/2022	12/01/2019	798,227	1.430%	1.430%
SLGS	Note	06/01/2023	12/01/2019	803,933	1.410%	1.410%
SLGS	Note	12/01/2023	12/01/2019	44,769,601	1.390%	1.390%
				51,298,130		

SLGS Summary

SLGS Rates File	29AUG19
Total Certificates of Indebtedness	1,793,555.00
Total Notes	49,504,575.00
Total original SLGS	51,298,130.00

Notes:

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3. Interest rate assumptions are based on current market conditions and similar credits.
4. Actual results may differ, and Stifel makes no commitments to underwriter at these levels.
5. Costs of issuance and underwriter's discount are estimates for discussion purposes.
6. Analysis was performed with no changes to the term or the structure of the debt service from the currently outstanding issue.
7. Non-interest paying capital appreciation bonds have been included in a manner consistent with the District's existing debt structure solely to generate premium to pay costs of issuance and fund a portion of the escrow to comply with State law requirements restricting the par amount of the refunding bonds.

ESCROW STATISTICS

Brecksville-Broadview Heights City School District Taxable Advance Refunding of School Facilities Improvement Bonds, Series 2018

Total Escrow Cost	Modified Duration (years)	Yield to Receipt Date	Yield to Disbursement Date	Perfect Escrow Cost	Value of Negative Arbitrage	Cost of Dead Time
Global Proceeds Escrow:						
51,298,130.22	3.716	1.397453%	1.397453%	47,908,796.57	3,389,333.59	0.06
51,298,130.22				47,908,796.57	3,389,333.59	0.06

Delivery date	10/15/2019
Arbitrage yield	3.249984%

Notes:

1. Preliminary and subject to change.
2. The use of the Aa2 rating is consistent with the rating of the outstanding prior bonds.
3. Interest rate assumptions are based on current market conditions and similar credits.
4. Actual results may differ, and Stifel makes no commitments to underwriter at these levels.
5. Costs of issuance and underwriter's discount are estimates for discussion purposes.
6. Analysis was performed with no changes to the term or the structure of the debt service from the currently outstanding issue.
7. Non-interest paying capital appreciation bonds have been included in a manner consistent with the District's existing debt structure solely to generate premium to pay costs of issuance and fund a portion of the escrow to comply with State law requirements restricting the par amount of the refunding bonds.

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